

Likhith Sanathi

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EDUCATION

University College Cork

Bachelor of Science in Computer Science (On track First-Class Honours - 1.1)

Cork, Ireland

Sep 2024 – May 2028

EXPERIENCE

IC Verification Intern

May 2026 – Sep 2026

Vishay Intertechnology, Inc.

Cork, Ireland

- Collaborated on UVM-based functional verification and regression workflows using vManager, supporting 100+ regression runs and gaining hands-on experience with industry-standard verification methodologies, automation flows, and regression management in a Linux environment.
- Leveraged AI tools to accelerate regression analysis, debugging, verification reporting, and license optimisation, reducing manual effort by 25–30% across large-scale workflows.
- Analysed regression, coverage, and traceability reports to identify verification gaps, improve functional coverage by approximately 10–15%, uncover missing test scenarios, and generate tailored reports for 3+ engineering disciplines.

PROJECTS

Autonomous Coding Harness | *Rust, Ratatui, OpenRouter, Crossterm*

May 2026 – July 2026

- Engineered a terminal-native autonomous coding runtime in Rust with a 6-stage ReAct-style plan-act-observe-reflect-verify loop for repository-aware software tasks.
- Built a structured agent execution layer with 21 typed tool actions for repo search, filesystem mutation, patching, archive handling, shell execution, and task finalization.
- Designed enterprise-style safety controls including reflection tracking, changed-file monitoring, approval gates, external-path isolation, timeouts, mutation-risk classification, and automated verification across 6 project types.

S&P-500 Quant Trading Pipeline | *Python, PyTorch, pandas, NumPy, CUDA*

Oct 2025 – Jan 2026

- Built a reproducible ML pipeline to forecast returns across 500+ large-cap equities, using 20+ technical indicators and 60-day rolling windows with robust OHLCV data cleaning and normalization.
- Implemented and trained 3 GPU-accelerated PyTorch sequence models (LSTM, GRU, Attention-LSTM) using chronological splits, Adam optimisation, LR scheduling, gradient clipping, and patience-based early stopping to reduce overfitting and improve generalization stability.
- Developed a backtesting engine with transaction costs, rule-based execution, and performance reporting across 10+ tickers; achieved 55.7% directional accuracy and outperformed buy-and-hold in sample AAPL backtests over the same evaluation horizon.

ACHIEVEMENTS

Winner, University of Galway CompSoc Intervarsity CTF (1/40 Teams)

Feb 2026

- Won Ireland's largest student-led Capture The Flag event, finishing 1st of 40 teams.

5th Place, ZeroDays CTF Open Division (16/153 Teams Globally)

Mar 2026

- Placed 5th overall in the Open category at one of the world's largest in-person Capture The Flag events.

PicoCTF Global Division: 81st / 8,747 Teams Worldwide (99.2nd Percentile)

Mar 2026

- Ranked in the top 0.8% worldwide in one of the largest global cybersecurity competitions.

UCC College of Science, Engineering and Food Science Undergraduate Scholarship

Sep 2024

- Awarded a University College Cork undergraduate scholarship from the College of Science, Engineering and Food Science.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, Rust, SQL, Verilog, SystemVerilog, Tcl, MIPS Assembly

Frameworks: Next.js, Express, Django, Flask, CUDA, PyTorch, UDP, TCP

Developer Tools: Git, Docker, Linux, Bash, AWS, Vercel, Verisium Manager, Integrated Metrics Centre (Cadence), VS Code, Visual Studio, PyCharm

Libraries: Pandas, NumPy, Matplotlib, Tailwind CSS, MaterialUI, ShadCN